UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

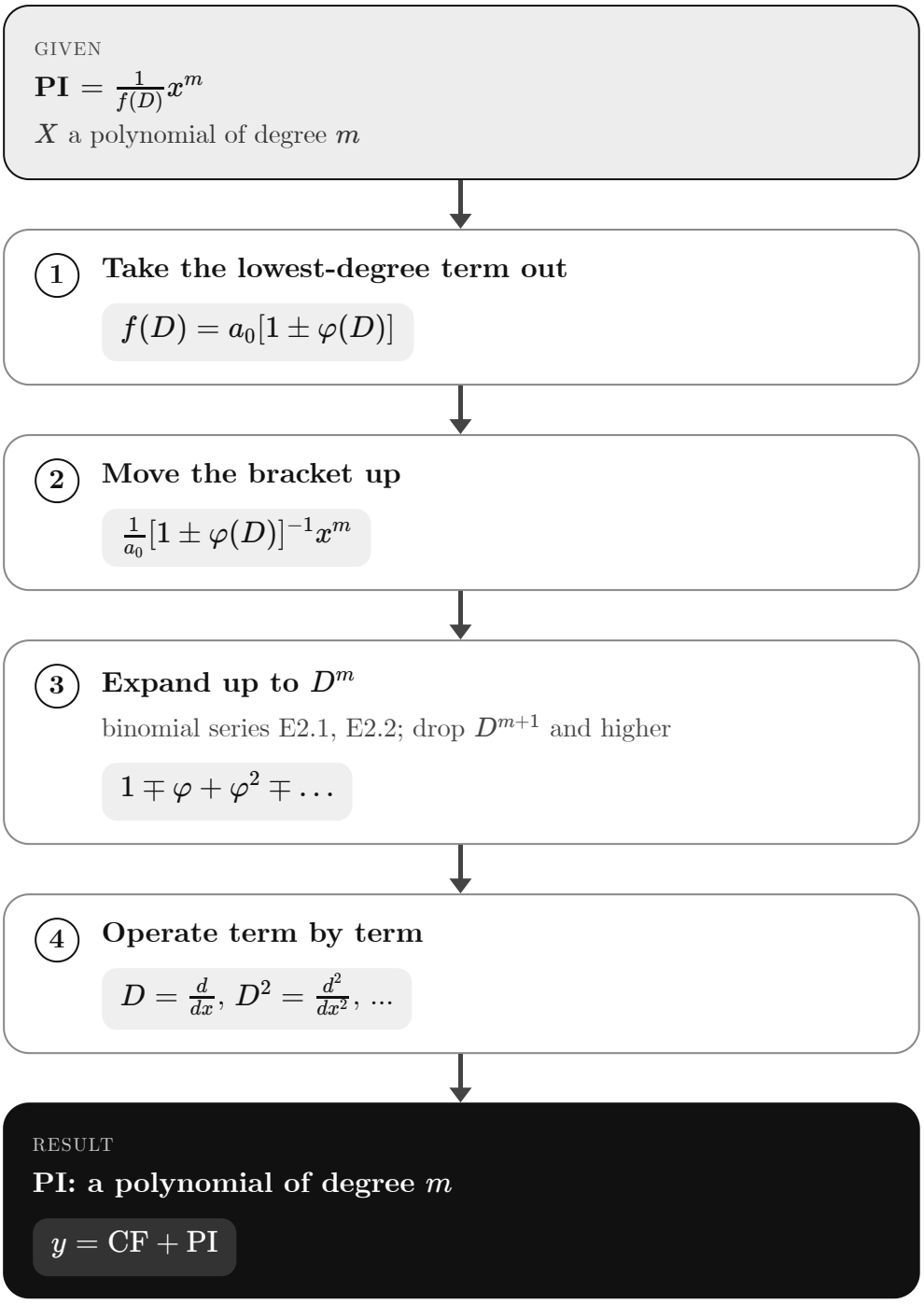
Class 005 · Particular integral Type 3 (X = x^m) · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S005·8	Unit 2 deck 005 (“Particular integral of Type 3”), slide 8 (rendered .pptx)
T005 00:06:48	Unit 2 AV-summary transcript “Particular integral of standard functions – Type 3” (12.6 min), segment start
T006a 00:06:05	Unit 2 AV-summary transcript “Particular integral – Type 3 and Type 4” (bucket 006, 15.5 min), its Type 3 part 00:00:00–00:09:55; the class 006 pack cites the whole file as T006b
CW U2 P15	Unit 2 class-work problem 15
ESA Jul-23 3b	Past paper: UE22MA141A Jul 2023, question 3.b
ESA Jan-24 s2 2d	Past paper: UE23MA141A Jan 2024 set 2, question 2.d

Contents

1. Figure F1.1 · The Type 3 procedure (slide 005·4)
2. Figure F2.1 · Which series?
3. Figure F2.2 · Which products survive? ($x^3 + x^2$, so $m = 3$)
4. Figure F3.1 · Constant term or factor D?
5. Figure F4.1 · Give each term of X its rule
6. Figure F4.2 · ESA Jul-23 3b: $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$
7. Figure F4.3 · ESA Jan-24 s1 2d: $(D^2 + 4)y = x^2 + \cos 2x + 2^{-x}$
8. Figure F4.4 · ESA Jan-24 s2 2d: $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$
9. Figure F4.5 · ESA Jan-24 s4 2c: $(D-1)^2(D+1)^2 y = \sin^2 \frac{x}{2} + 2^{-x} + x$
10. Figure F4.6 · CW P21: $(D-2)^2 y = 8(e^{2x} + \sin 2x + x^2)$
11. Types 1, 2 and 3 side by side
12. The four expansions of slide 005·5
13. Every mixed question with a polynomial term

1 The Type 3 rule: $X = x^m$



Remark. Collect the operator $(1 + c_1D + \cdots + c_mD^m)$ before differentiating.

Exam trap. Keep the factor $\frac{1}{a_0}$ and its sign ($D^2 - 1 = -(1 - D^2)$).

Figure F1.1. The Type 3 procedure (slide 005·4) ESA ★★★

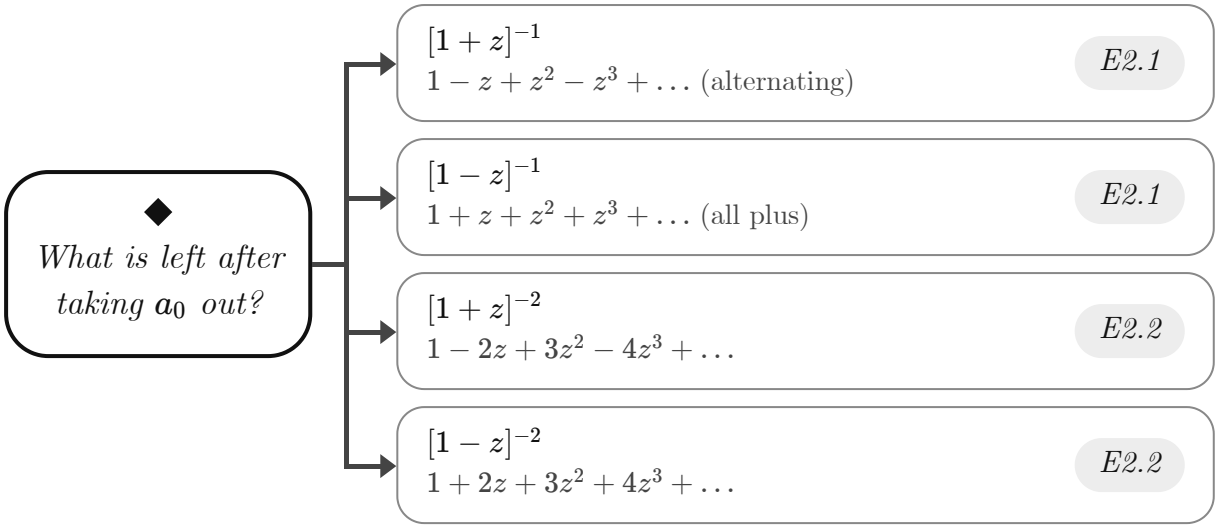
S005·4 · S005·10 · T005 00:01:08 · T005 00:02:17 · T006a 00:00:55

Types 1, 2 and 3 side by side

	Type 1: e^{ax}	**Type 2:** $\sin ax$, $\cos ax$	**Type 3:** x^m
What to do with D	replace D by a	replace D^2 by $-a^2$	expand $[f(D)]^{-1}$ in powers of D
PI	$\frac{e^{ax}}{f(a)}$	$\frac{\sin ax}{f(-a^2)}$	$\frac{1}{a_0}[1 \pm \varphi(D)]^{-1}x^m$
Special case	$f(a)=0$: $x\frac{e^{ax}}{f'(a)}$	$f(-a^2)=0$: $x\frac{\sin ax}{f'(-a^2)}$	$f(0)=0$: take D^k out, integrate
PI looks like	a multiple of e^{ax}	$A\cos ax + B\sin ax$	a polynomial of degree m (or $m+k$)
Class pack	003	004	005 (this pack)

added for clarity S005·4 · S005·10

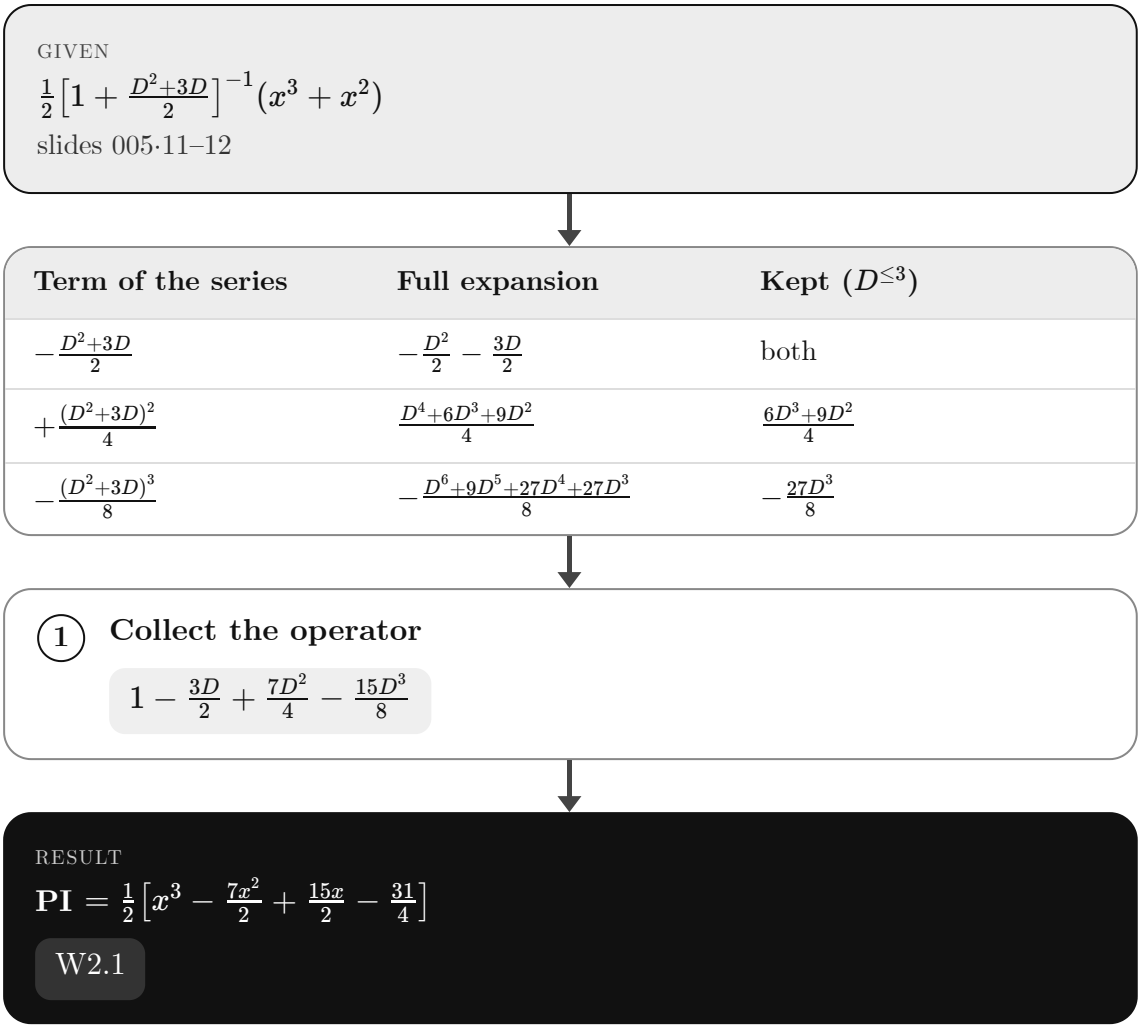
2 The binomial expansions and where to stop



Remark. $z = \varphi(D)$. Remember the first and third rows; the others follow by $z \to -z$.

Figure F2.1. Which series? ESA ★★★

S005·5 · T005 00:02:45 · T005 00:04:04



Remark. The fourth power and beyond are not written at all: every product in them has D^4 or more.

Figure F2.2. Which products survive? (x^3+x^2, so $m=3$) ESA ★★★

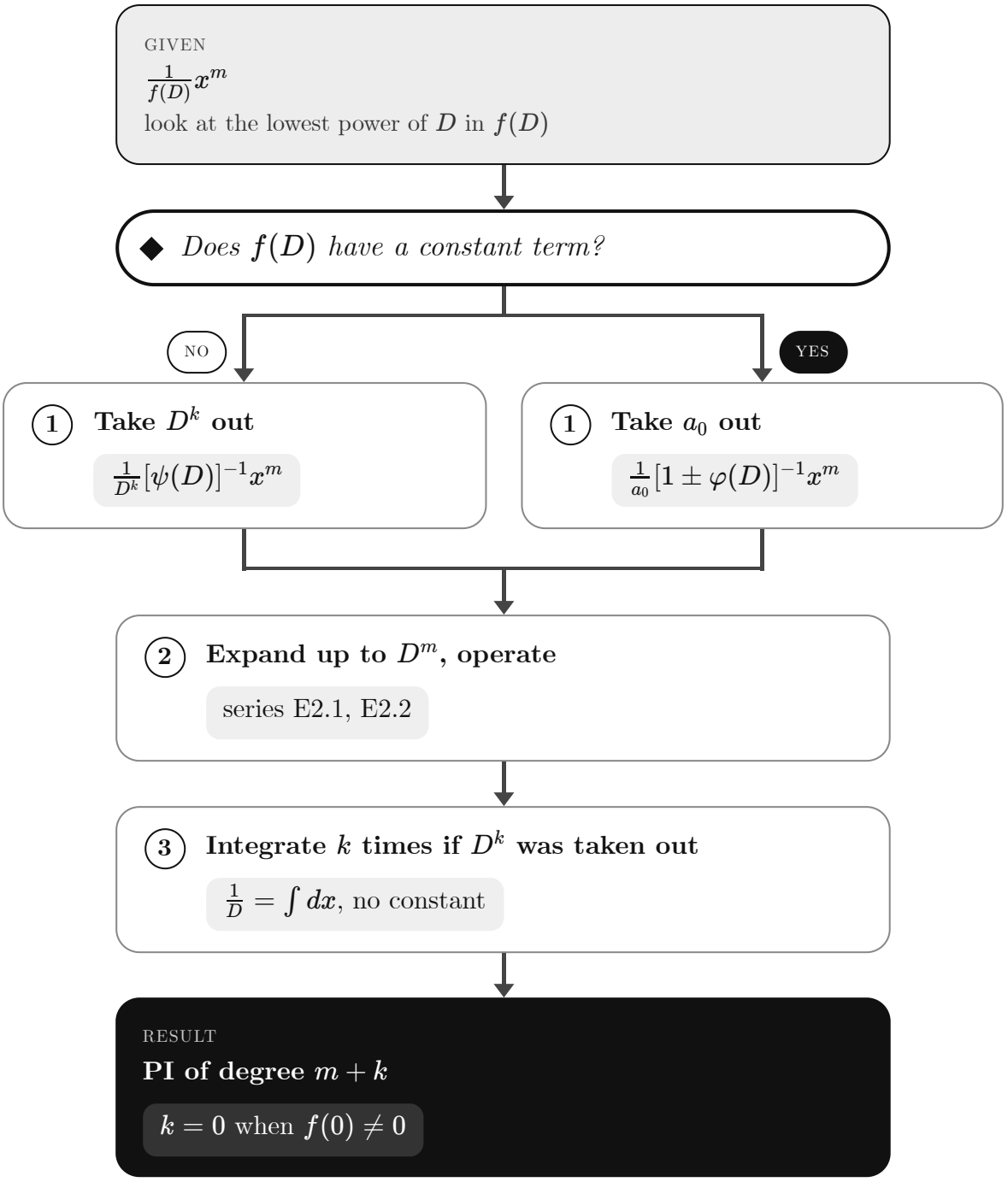
S005·12 · T006a 00:06:05 · T006a 00:06:43

The four expansions of slide 005·5

Expression	Series	Signs	Remember?
$(1+z)^{-1}$	$1-z+z^2-z^3+\dots$	alternate	yes
$(1-z)^{-1}$	$1+z+z^2+z^3+\dots$	all plus	from row 1, $z\to -z$
$(1+z)^{-2}$	$1-2z+3z^2-4z^3+\dots$	alternate	yes
$(1-z)^{-2}$	$1+2z+3z^2+4z^3+\dots$	all plus	from row 3, $z\to -z$

S005·5 · T005 00:03:03 · T005 00:04:04

3 When D is a factor of $f(D)$



Remark. $f(0) = 0$ means $m = 0$ is an AE root: the CF already contains a constant.

Figure F3.1. Constant term or factor D? ESA ★

ADDED FOR CLARITY S005·4

4 Mixed right-hand sides: Types 1, 2 and 3 together

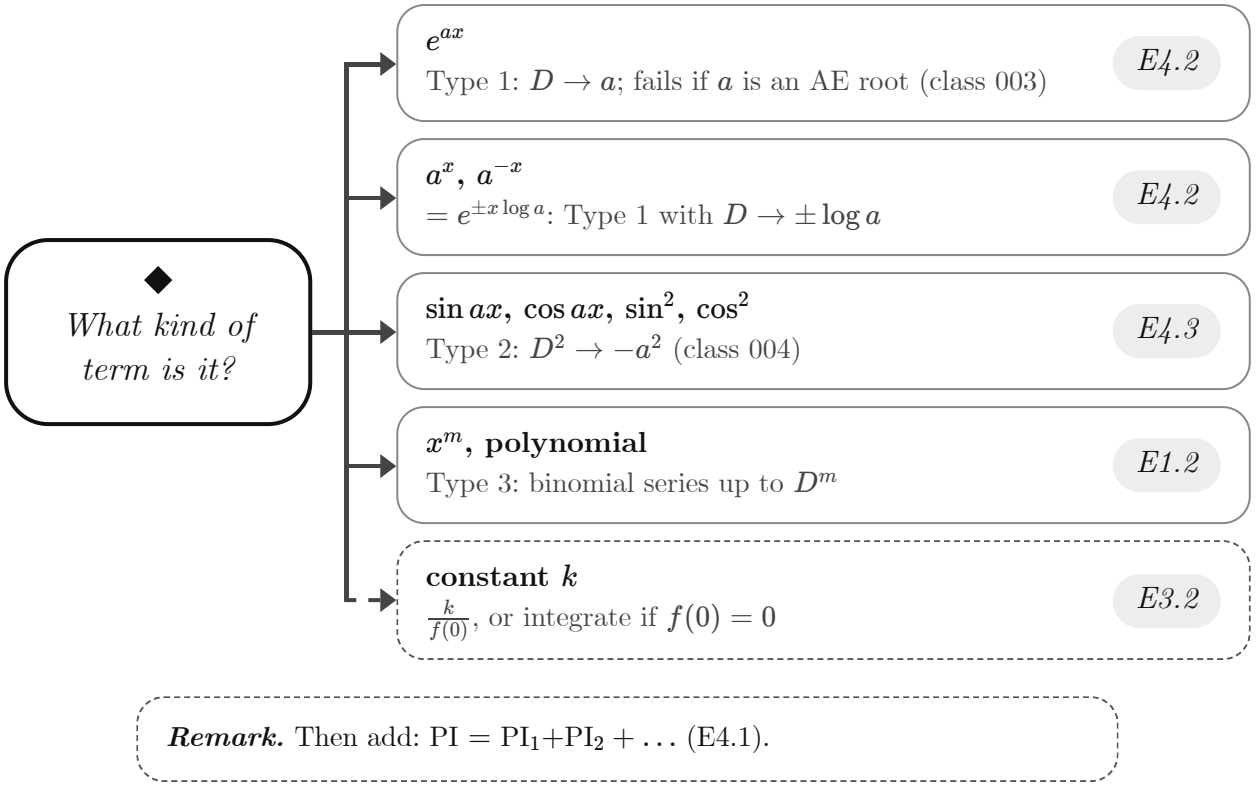
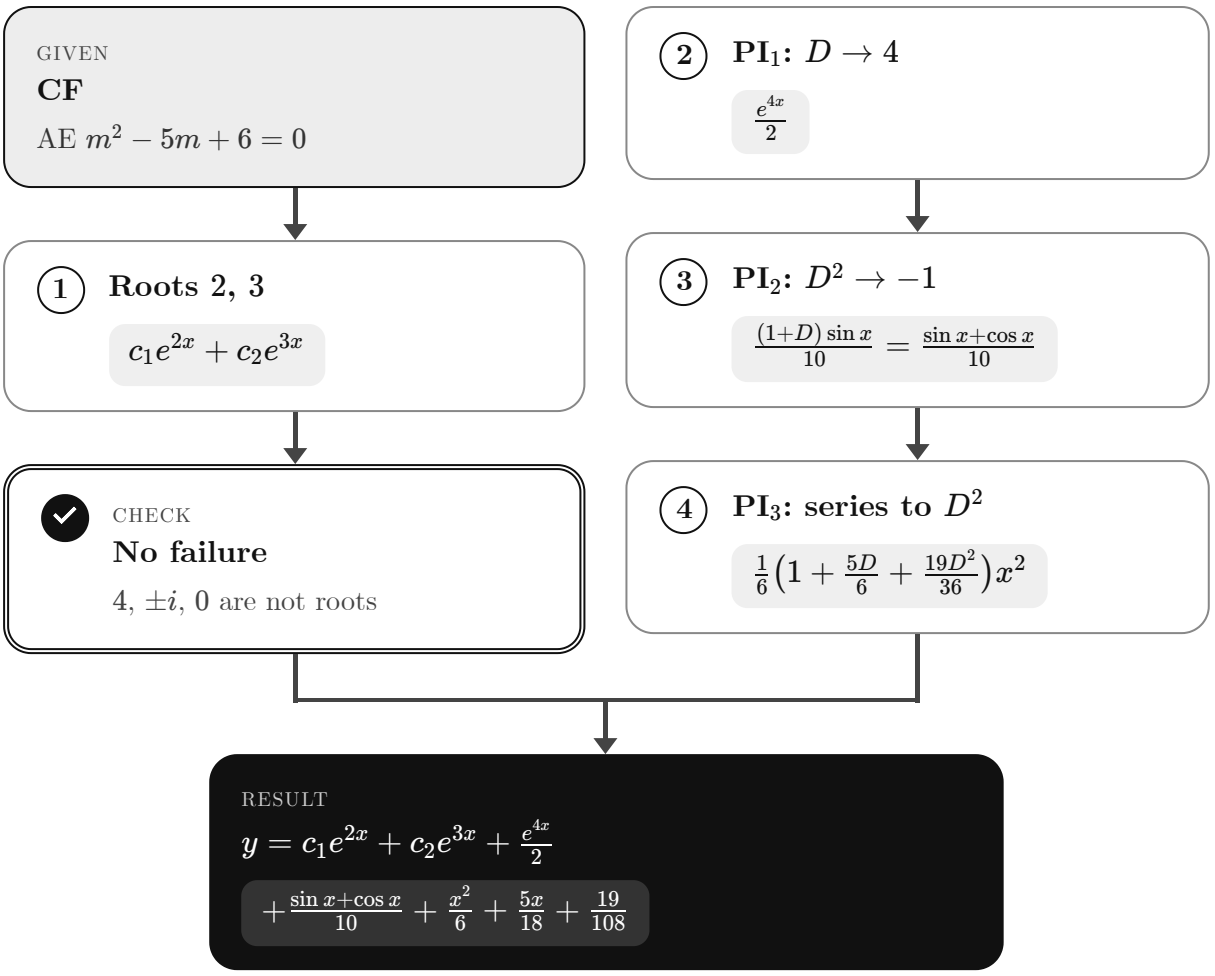


Figure F4.1. Give each term of X its rule ESA ★★★

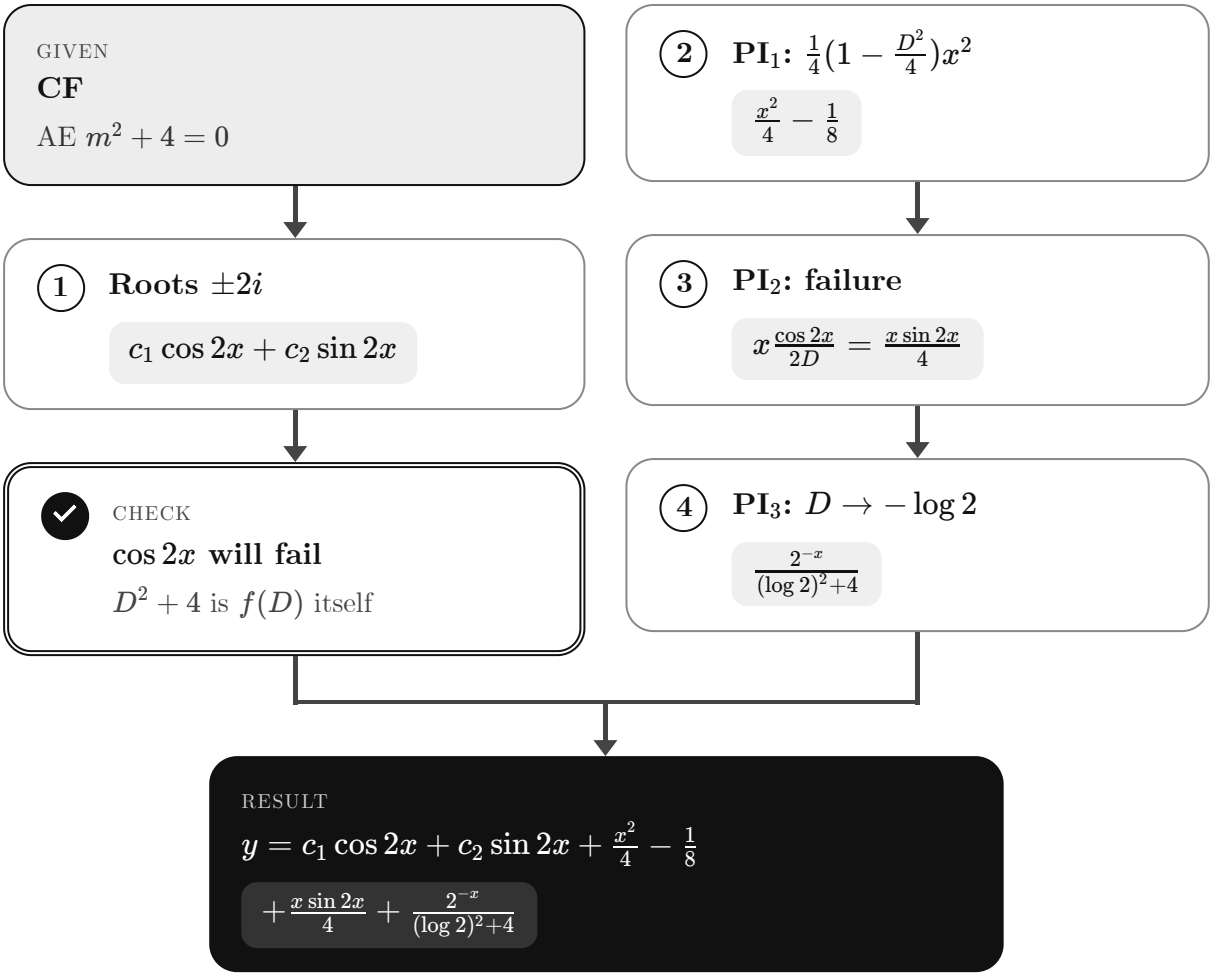
ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c



Remark. Worked in full as W4.1 (7 marks).

Figure F4.2. ESA Jul-23 3b: $(D^2 - 5D + 6)y = e^{4x} + \sin x + x^2$ ESA ★★★

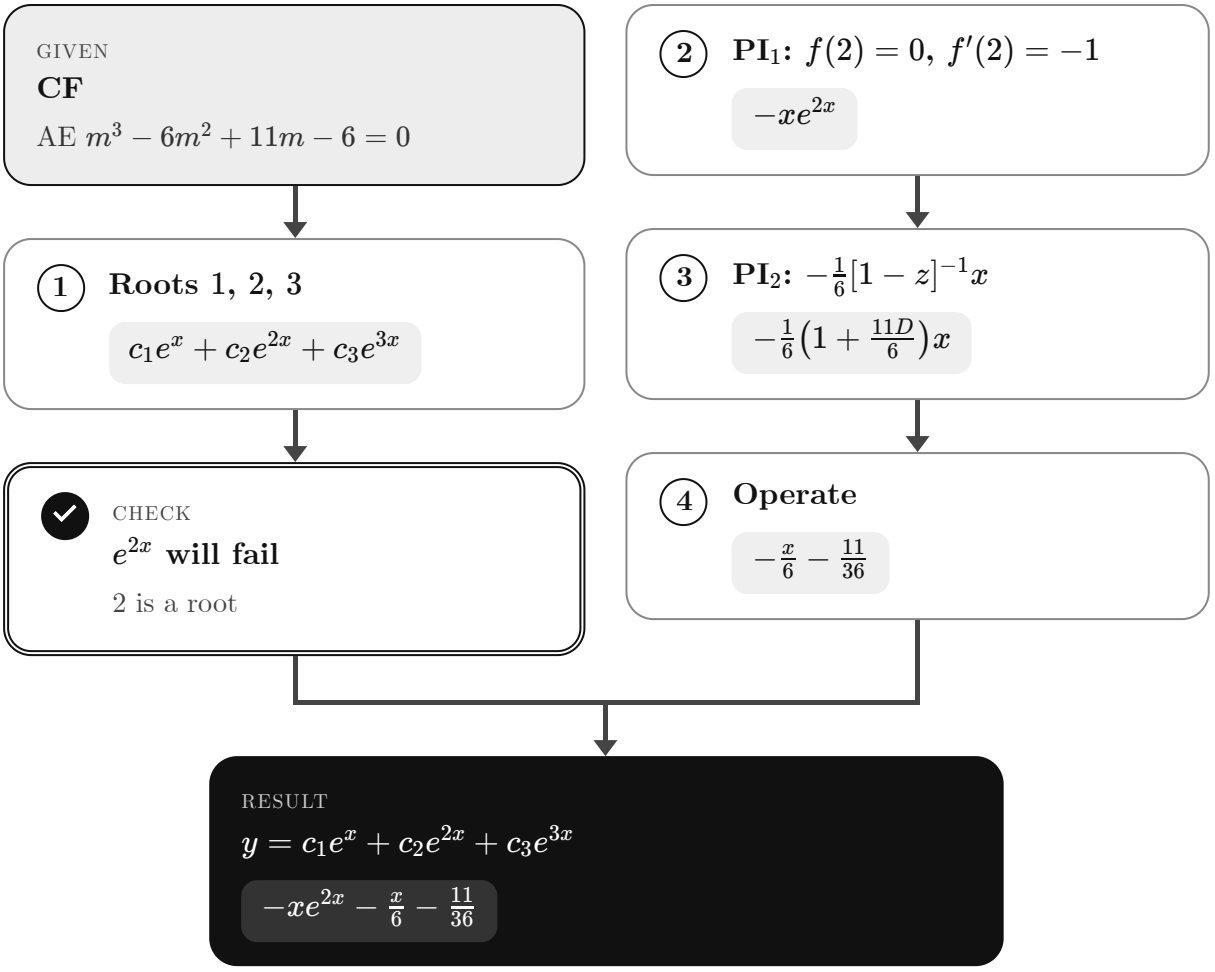
ESA Jul-23 3b



Remark. Worked in full as W4.2 (5 marks).

Figure F4.3. ESA Jan-24 s1 2d: $(D^2+4)y = x^2 + \cos 2x + 2^{-x}$ ESA ★★★

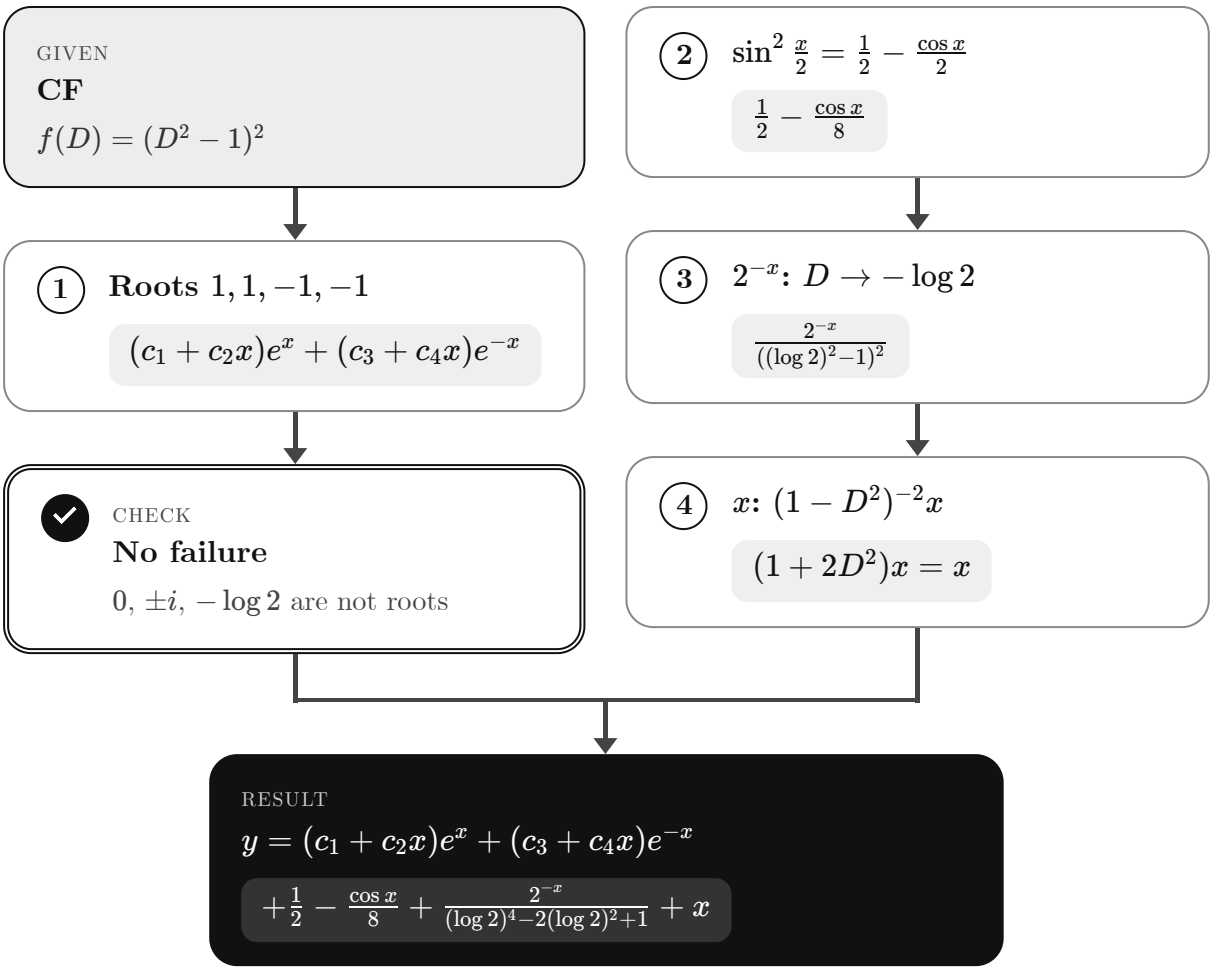
ESA Jan-24 s1 2d



Remark. For $X = x$ ($m = 1$) only $1 + z$ of the series is needed, and of z only its D term. Worked as W4.3.

Figure F4.4. ESA Jan-24 s2 2d: $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$ ESA ★★★

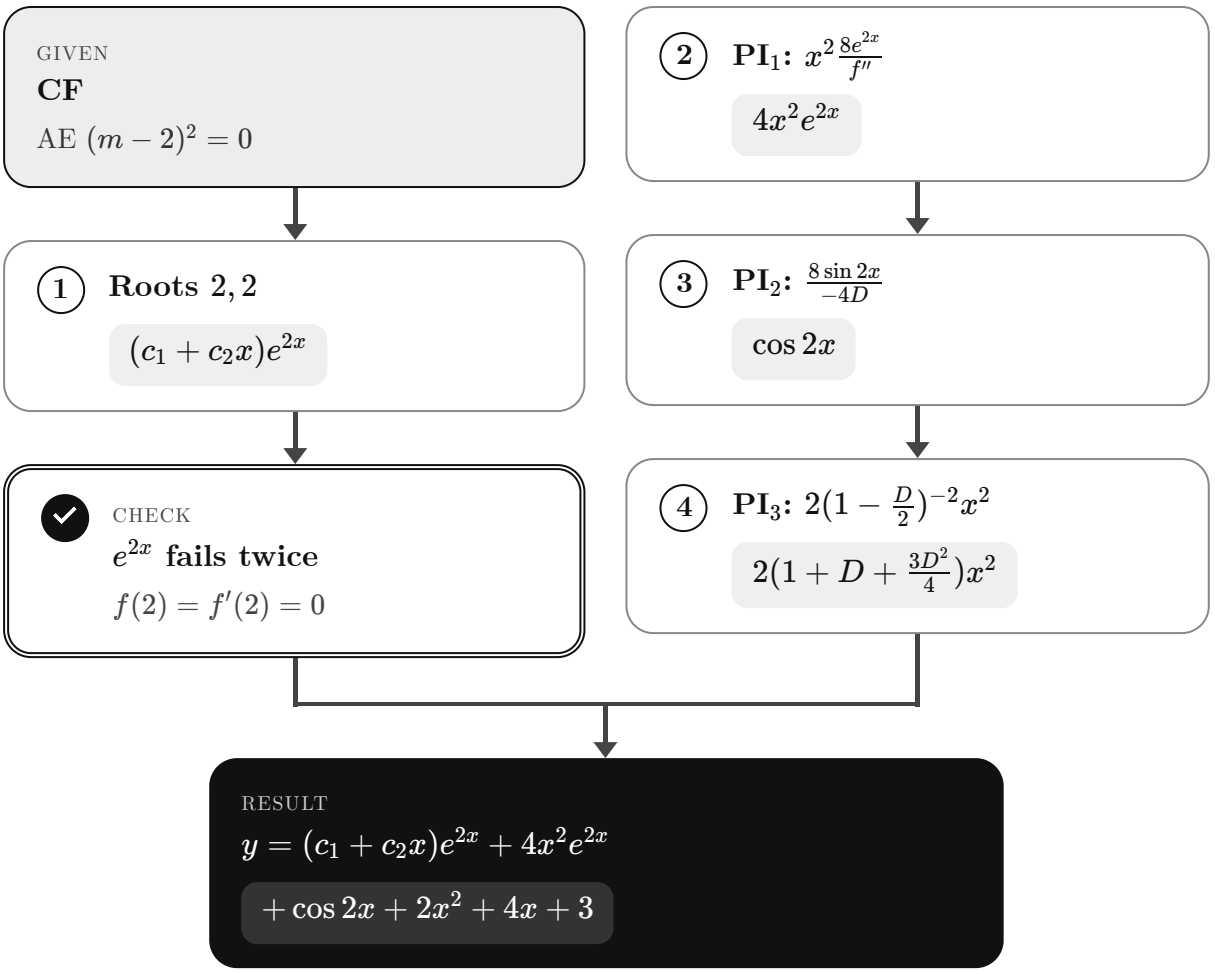
ESA Jan-24 s2 2d



Remark. Four terms, three rules. Worked in full as W4.4 (6 marks).

Figure F4.5. ESA Jan-24 s4 2c: $(D-1)^2(D+1)^2y = \sin^2\frac{x}{2} + 2^{-x} + x$
ESA ★★★

ESA Jan-24 s4 2c · CW U2 P22



Remark. The ESA question Dec-23 3a (asked three times) is this without x^2. Worked as W4.5.

Figure F4.6. CW P21: $(D-2)^2y=8(e^{2x}+\sin 2x+x^2)$ ESA ★★★

CW U2 P21 · ESA Dec-23 3a

Every mixed question with a polynomial term

Paper, Q	$f(D)$	Other terms (rule)	Polynomial term → PI	Worked
Jul-23 3b	$D^2 - 5D + 6$	e^{4x} (T1), $\sin x$ (T2)	$x^2 \to \frac{x^2}{6} + \frac{5x}{18} + \frac{19}{108}$	W4.1
Jan-24 s1 2d	$D^2 + 4$	$\cos 2x$ (T2, fails), 2^{-x} (T1)	$x^2 \to \frac{x^2}{4} - \frac{1}{8}$	W4.2
Jan-24 s2 2d	$D^3 - 6D^2 + 11D - 6$	e^{2x} (T1, fails)	$x \to -\frac{x}{6} - \frac{11}{36}$	W4.3
Jan-24 s4 2c (= CW P22)	$(D-1)^2(D+1)^2$	$\sin^2 \frac{x}{2}$ (T2 + const), 2^{-x} (T1)	$x \to x$	W4.4
CW P21	$(D-2)^2$	$8e^{2x}$ (T1, fails twice), $8\sin 2x$ (T2)	$8x^2 \to 2x^2 + 4x + 3$	W4.5

ADDED FOR CLARITY ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s4 2c · CW U2 P21